

ABERDARE TECHNICAL COLLEGE
INFORMATION COMMUNICATION TECHNOLOGY
MODULE 2, TERM 2
PRACTICAL EXAMINATION 1
UNIT OF COMPETENCY: NETWORK SETUP
UNIT CODE: 0612 351 03A
LEVEL: 5
ELEMENT 1: TERMINATE NETWORK CABLES
TIME: 2 HOURS

MATERIALS

- ✓ UTP Cat 5e/Cat 6 network cable
- ✓ RJ45 connectors
- ✓ RJ45 crimping tool
- ✓ Cable stripper
- ✓ Cable tester
- ✓ Cable labels or marker pen

INSTRUCTIONS TO THE CANDIDATE

- ✓ *Read all instructions carefully before commencing the practical task.*
- ✓ *Observe all workshop safety requirements.*
- ✓ *Use the tools and materials provided.*
- ✓ *Terminate one straight-through cable and one crossover cable.*
- ✓ *Use the correct T568A and T568B wiring standards.*
- ✓ *Ensure that the conductors are arranged in the correct sequence before inserting them into the RJ45 connector.*
- ✓ *Ensure that the conductors reach the end of the RJ45 connector before crimping.*
- ✓ *Test the completed cables using a cable tester.*
- ✓ *Correct any faults identified during testing.*
- ✓ *Clearly label the completed cables as **STRAIGHT-THROUGH** and **CROSSOVER**.*
- ✓ *Maintain a clean and organised workstation throughout the practical.*

- ✓ *Return all tools and unused materials after completing the task.*

PRACTICAL TASK

You have been provided with UTP cable, RJ45 connectors, a crimping tool, a cable stripper and a cable tester.

Perform the following tasks.

TASK 1: TERMINATE A STRAIGHT-THROUGH CABLE

- ✓ Terminate a straight-through Ethernet cable using:
- ✓ **T568B on both ends**
- ✓ Use the correct color codes T568B Color sequence:
- ✓ After termination, test the cable using the cable tester.

TASK 2: TERMINATE A Crossover CABLE

Terminate a crossover Ethernet cable using:

T568A on one end → T568B on the other end

After termination, test the cable using the cable tester.

TASK 3: TEST AND PRESENT THE CABLES

1. *Connect the straight-through cable to the cable tester.*
2. *Observe the tester sequence.*
3. *Record the test result.*
4. *Connect the crossover cable to the cable tester.*
5. *Observe the tester sequence.*
6. *Record the test result.*
7. *Identify and correct any termination faults.*
8. *Label both completed cables appropriately.*
9. *Present the completed cables to the assessor.*

The trainer is introducing oral questions shortly